Most of these results were forgotten. But these old results recently have come roaring back.

4.2 VOLATILITY ANOMALY

I was fortunate to write one paper that helped launch the new "risk anomaly" literature in 2006 with Robert Hodrick, one of my colleagues at Columbia Business School, and two of our former students, Yuhang Xing and Xiaoyan Zhang, who are now professors at Rice University and Purdue University, respectively. We found that the returns of high-volatility stocks were "abysmally low." So low that they had zero average returns. This paper now generates the most cites per year of all my papers and has spawned a follow-up literature attempting to replicate, explain, and refute the results.[24]

First, should there even be a relation between volatility and returns? The whole point of the CAPM and the many multifactor extensions (see chapter 7) was that stock return volatility itself should not matter. Expected returns, according to these models, are determined by how assets covary with factor risks. Idiosyncratic volatility, or tracking error (see equation (10.3)), should definitely *not* have any relation to expected returns under the CAPM. But in markets that are segmented due to clientele effects—where some agents cannot diversify or where some agents prefer to hold some assets over others for exogenous reasons—idiosyncratic volatility should be positively related to returns. Intuitively, agents have to be paid for bearing idiosyncratic risk, resulting in a positive relation between idiosyncratic risk and volatility in equilibrium. In later models with "noise traders," who trade for random reasons unrelated to fundamental valuation, higher volatilities are associated with higher risk premiums.[25]

The Ang et al. (2006) results show exactly the opposite.

Particularly notable is the robustness of the negative relation between both idiosyncratic and total volatility with returns. We employed a large number of controls for size, value, leverage, liquidity risk, volume, turnover, bid–ask spreads, co-skewness risk, dispersion in analysts' forecasts, and momentum. We also did not find that aggregate volatility risk explained our result—even though volatility risk is a pervasive risk factor (see chapter 7). In subsequent work, Ang et al. (2009), we showed that the volatility effect existed in each G7 country and across all developed stock markets. We also controlled for private information, transactions costs, analyst coverage, institutional ownership, and delay measures, which

[24] Volatility makes many appearances, of course, in tests of cross-sectional asset pricing models before Ang et al. (2006), but most of them are negative results or show a slight positive relation. For example, in Fama and MacBeth's (1973) seminal test of the CAPM, volatility is included and carries an insignificant coefficient. Eric Falkenstein (2012) recounts that he uncovered a negative relation between volatility and stock returns in his PhD dissertation in 1994, which was never published.

[25] For clientele models, see Merton (1987). For noise trader models, see DeLong et al. (1990).